

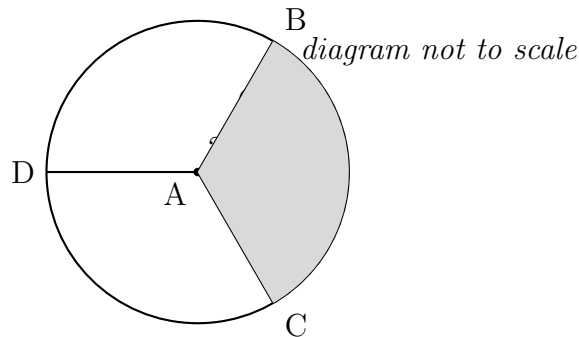
IB Mathematics Standard Level

Paper 1

12 November 2018 – No Calculator
90 marks total

Section A

1. [6 marks] The following diagram shows a circle with centre A and radius 6 cm.



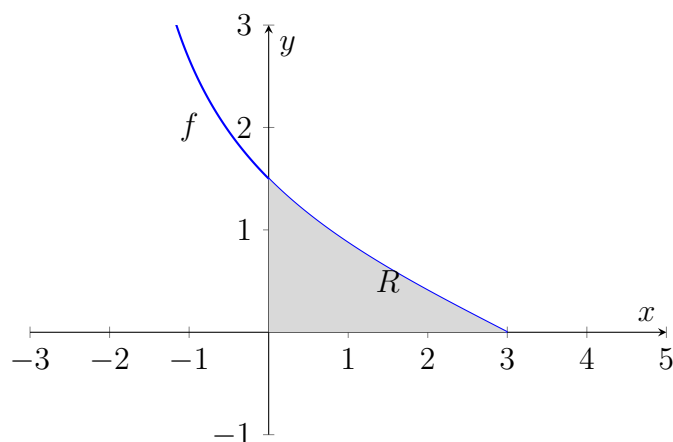
The points B, C, and D lie on the circle, and $\widehat{BAC} = 2$ radians.

- (a) Find the area of the shaded sector.
- (b) Find the perimeter of the non-shaded sector ABDC.
2. [5 marks] Two functions, f and g , are defined in the following table.

x	-2	1	3	6
$f(x)$	6	3	1	-2
$g(x)$	-7	-2	5	9

- (a) Write down the value of $f(1)$.
- (b) Find the value of $(g \circ f)(1)$.
- (c) Find the value of $g^{-1}(-2)$.
3. [6 marks] In an arithmetic sequence, $u_1 = -5$ and $d = 3$.
- (a) Find u_8 .

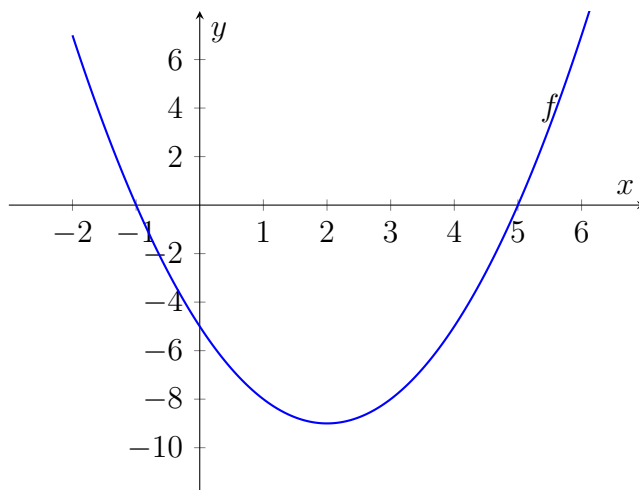
- (b) Find the value of n for which $u_n = 67$.
4. **[6 marks]** Let $b = \log_2 a$, where $a > 0$. Write down each of the following expressions in terms of b .
- (a) $\log_2 a^3$
 (b) $\log_2 8a$
 (c) $\log_8 a$
5. **[6 marks]** Consider the vectors $\mathbf{a} = \begin{pmatrix} 3 \\ 2p \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} p+1 \\ 8 \end{pmatrix}$.
 Find the possible values of p for which $\mathbf{a}$ and $\mathbf{b}$ are parallel.
6. **[8 marks]** Let $f(x) = \frac{6-2x}{\sqrt{16+6x-x^2}}$. The following diagram shows part of the graph of f .



- The region R is enclosed by the graph of f , the x -axis, and the y -axis. Find the area of R .
7. **[6 marks]** Given that $\sin x = \frac{1}{3}$, where $0 < x < \frac{\pi}{2}$, find the value of $\cos 4x$.

Section B

8. [16 marks] Let $f(x) = x^2 - 4x - 5$. The following diagram shows part of the graph of f .



- (a) Find the x -intercepts of the graph of f .
- (b) Find the equation of the axis of symmetry of the graph of f .
- (c) The function can be written in the form $f(x) = (x - h)^2 + k$.
 - (i) Write down the value of h .
 - (ii) Find the value of k .

The graph of a second function, g , is obtained by a reflection of the graph of f in the y -axis, followed by a translation of $\begin{pmatrix} -3 \\ 6 \end{pmatrix}$.

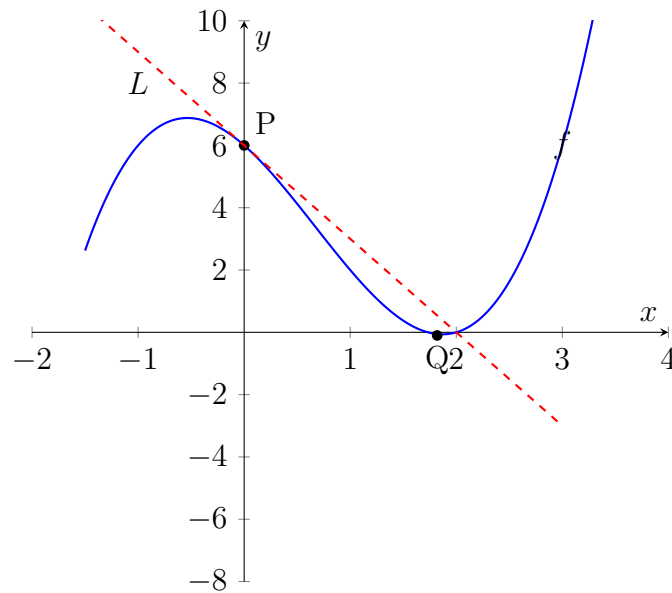
- (d) Find the coordinates of the vertex of the graph of g .
9. [15 marks] A bag contains n marbles, two of which are blue. Hayley plays a game in which she randomly draws marbles out of the bag, one after another, without replacement. The game ends when Hayley draws a blue marble.
- (a) Find the probability, in terms of n , that the game will end on her
 - (i) first draw;
 - (ii) second draw.
 - (b) Let $n = 5$. Find the probability that the game will end on her
 - (i) third draw;
 - (ii) fourth draw.

Hayley plays the game when $n = 5$. She pays \$20 to play and can earn money back depending on the number of draws it takes to obtain a blue marble. She earns no money back if she obtains a blue marble on her first draw. Let M be the amount of money that she earns back playing the game. This information is shown in the following table.

Number of draws	1	2	3	4
Money earned back (\$ M)	0	20	$8k$	$12k$

(c) Find the value of k so that this is a fair game.

10. **[16 marks]** Let $f(x) = x^3 - 2x^2 + ax + 6$. Part of the graph of f is shown in the following diagram.



The graph of f crosses the y -axis at the point P. The line L is tangent to the graph of f at P.

- (a) Find the coordinates of P.
- (b)
 - (i) Find $f'(x)$.
 - (ii) Hence, find the equation of L in terms of a .

The graph of f has a local minimum at the point Q. The line L passes through Q.

- (c) Find the value of a .